

Gábor Hosu

contact@hgabor.com | +40 74 252 6970 | hgabor.com | github.com/gabor-hosu | linkedin.com/in/gabor-hosu

Honours and Publications

- Á. Kránics, **G. Hosu**, et al., “Multimodal RAG for Context-Aware Chatbots”,
IEEE SISY 23rd [DOI], [Preprint] Oct 2025
- 2nd Prize** at 28th Transylvanian Students’ Scientific Conference May 2025

Experience

- Software Engineering Internship | Codespring** Jul 2024 - Sept 2024
- Designed and implemented an **end-to-end backend pipeline** for a *[RAG chatbot]*, handling **text and image file ingestion, preprocessing, vector embedding, and retrieval**.
 - Developed **REST API endpoints** with **custom authentication** and **chat, message, and file management**.
 - Containerized services** with Docker and implemented a **GitLab CI/CD pipeline** with code linting, unit tests, **automatically deploying the RAG service on an internal server** and the **main backend to Azure free tier**.
 - Collaborated in a **2-person Scrum team**, contributing to sprint planning, development, and **remote development/mentoring during university**, which continued into a **BSc thesis** and **IEEE publication**.
 - Technologies:** Git, GitLab, Docker, Azure, MS SQL, Milvus, Python (FastAPI, SQLAlchemy, LangChain)
- Fullstack mentorprogram | Codespring** Oct 2023 - May 2024
- Participated in a **pre-internship professional training program**, consisting of **2 sessions per month**
 - Gained hands-on experience with **fullstack technologies** and practical software development concepts.
 - Technologies:** Java, Spring, React

Projects

- Embedding-Based Retrieval Paper Reproduction** Nov 2025 - Jan 2026
[Portfolio] | [GitHub] | [Paper] | [HF (Models & Data)]
- Reproduced key results of a **2025 Google DeepMind paper** on geometric limits of single-vector embedding retrieval under **GPU constraints**.
 - Generated synthetic **IR datasets (2K–50K docs)** and **quantized models**; published artifacts on Hugging Face.
 - Implemented embedding optimization and **benchmarking pipelines** with **quantized LLMs**.
 - Fine-tuned** and implemented the full **model pipeline**, experimenting with **datasets** and **hyperparameters**.
 - Presented findings in an **academic seminar** and as a final Information Retrieval course project.
 - Technologies:** Python, PyTorch, NumPy, Pandas, Sentence-Transformers, Hugging Face, Milvus, CUDA, Linux

Education

- Babeş-Bolyai University**, MSc, Data Analysis and Modelling Oct 2025 - Present
- Relevant coursework:** Information Retrieval, Swarm Intelligence, Robotics
 - Team projects:** Contributed to **YOLOv8 model training and deployment** on live Cozmo robot video feed; collaborated on **truss design optimization** with PSO and FEM simulations.
 - Delivered a *[presentation]* about **Milvus vector database**, analyzing its implementation details.
- Babeş-Bolyai University**, BSc, Mathematics and Computer Science 2022 - 2025

Languages

English: Intermediate/Upper-Intermediate
Romanian: Intermediate
Hungarian: Native